Society of Computer Science and Engineering

XAVENIR – 2026 | AI/ML Challenge

RULES:

1. **Problem Statement 1 is open to all students including 1st years.**
2. **Problem Statement 2 is only for 1st-year B.Tech. students.**
3. **Problem Statement 1 is eligible for overall prizes.**
4. **Problem Statement 2 has a separate exclusive prize for 1st-year participants.**
5. **Participants are free to use any resources, datasets, or tools —**

   EXCEPT pretrained model backbones or pretrained embeddings of any kind.

   (No BERT, RoBERTa, GPT, Word2Vec, GloVe, FastText, or any pretrained weights. All representations must be learned from scratch.)
6. **Final submission must include the complete code and a PPT covering:**

   Approach, Methodology, Results, and Evaluation Metrics.

PROBLEM STATEMENT 1 (Open to All Years):

"Catch the Liar" — Hallucination Detection in AI-Generated Text

Background:

Large Language Models are increasingly used to summarize scientific content. However, these models can be manipulated — causing them to hallucinate, drift off-topic, introduce factual errors, or produce outputs subtly inconsistent with the source. In this challenge, you are given pairs of AI-generated summaries of the same article. One is from a clean model, the other from a manipulated one. Your job is to find the liar.

What You Need to Do:

Given a pair of summaries (Text A, Text B) of the same source article, output:

- A label: 1 if Text A is the hallucinated output, 2 if Text B is.

No pretrained backbones. Every feature and representation must be built from scratch on the provided data.

Approaches to Explore:

- Linguistic features: vocabulary richness, sentence length variance, POS tag ratios, readability scores, Unicode character ratios.
- Statistical divergence between the two texts (e.g., KL divergence over word distributions).
- Train your own tokenizer and embeddings from scratch, then build a simple classifier (Naive Bayes, SVM, shallow MLP).
- Consistency checks: does the summary stay on topic? Are named entities from the source preserved?

Dataset:

- ESO "The Messenger" journal articles + LLM-generated summary pairs.
- https://www.kaggle.com/competitions/fake-or-real-the-impostor-hunt/data
- Training set has ground truth labels. Test set does not.

Evaluation Criteria:

- Accuracy on test set (primary metric)
- Novelty and clarity of approach
- Simplicity vs. performance tradeoff

References:

[1] Kaczmarek et al., "Fake or Real: The Impostor Hunt in Texts for Space Operations", arXiv:2507.13508v3, 2025.

[2] Sriramanan et al., "LLM-Check: Investigating Detection of Hallucinations

in Large Language Models", NeurIPS 2024, pp. 34188-34216.
[3] Yehuda et al., "InterrogateLLM: Zero-Resource Hallucination Detection in LLM-Generated Answers", 2024.

PROBLEM STATEMENT 2 (Only for 1st-year B.Tech and MCA): "The Turing Filter" — Human-Written vs. AI-Generated Text Detection

Background:

Modern LLMs produce fluent, coherent text that superficially resembles human writing. Yet subtle statistical and linguistic signatures often betray their origin. In this challenge, you are given text samples — some written by humans, some generated by LLMs. Build a classifier that tells them apart. No pretrained models allowed.

What You Need to Do:

For each input text, output:

- A binary label: 0 for human-written, 1 for AI-generated.

Approaches to Explore:

- Feature engineering: word length, sentence variance, punctuation patterns, type-token ratio.
- N-gram language model trained on the corpus to estimate perplexity.
- Classical ML: TF-IDF + logistic regression / SVM / random forest.
- Small RNN or LSTM trained from scratch on the provided data.

Dataset:

- https://www.kaggle.com/competitions/llm-detect-ai-generated-text/data
- Student-written essays vs. LLM-generated essays, with training labels.

Evaluation Criteria:

- Accuracy, Precision, Recall, F1 on test set
- Creativity in feature design
- Reproducibility and clarity of pipeline

References:

[1] Kaczmarek et al., "Fake or Real: The Impostor Hunt in Texts for Space Operations", arXiv:2507.13508v3, 2025.

[2] Wang & Chaudhuri, "Data Poisoning Attacks Against Online Learning", 2018.

[3] Chen et al., "Targeted Backdoor Attacks on Deep Learning Systems Using Data Poisoning", 2017.

In case of any queries, feel free to contact the Event Heads:

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